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RESEARCH INTEREST

My research focuses on multimodal clinical AI, particularly how AI-derived clinical variables relate to patients' underlying clinical states and to the healthcare processes that generate the data from which those variables are derived. I am interested in developing methods to characterize these measurement relationships across clinical text, medical imaging, physiological data, and patient-reported outcomes, including when they change with disease or symptom severity. More broadly, my work spans longitudinal health data, large language models, multimodal learning, model generalization, fairness, and clinically deployed AI systems.

EDUCATION

Vanderbilt University — Nashville, TN, USA
M.S. in Electrical Engineering and Computer Science — *Aug 2018 – Dec 2019*

- GPA: 3.93/4.0
- Advised by: Prof. Alan Peters
- Research: Computer vision for stereo 3D reconstruction, and CNNs for object recognition in images and videos

Rajiv Gandhi Technological University — Indore, MP, India
B.S. in Electronics and Communications — *Aug 2013 – May 2017*

- GPA: 3.5/4.0
- Advised by: Dr. D.K. Panda
- Research: Home automation using internet of things (IoT)

PUBLICATIONS

Peer-Reviewed Journal Articles (* *denotes equal contribution;* † *denotes corresponding author*)

[J1] ***Thakkar V****, *Silverman GM**, *Kc A, Ingraham NE, Jones EK, King S, Melton GB, Zhang R, Tignanelli CJ*. **A Comparative Analysis of Large Language Models versus Traditional Information Extraction Methods for Real-World Evidence of Patient Symptomatology in Acute and Post-Acute Sequelae of SARS-CoV-2.** *PLOS ONE*. 2025;20(5):e0323535.

[J2] ***Thakkar V†,*** *Silverman GM, Putzer M, Kc A, Ikramuddin S, Melnik T, Ingraham NE, Keller N, Jones EK, King S, Bailey A, Melton GB, Tignanelli CJ*. **Longitudinal Symptom Severity Tracking in Vagus Nerve Stimulation Patients: A Two-Stage LLM-Based Pipeline with Explainable AI.** *JAMIA Open*. Accepted September 2026.

[J3] ***Thakkar V†****, Phillips M, Loftus TJ, Burns JL, Switalla KM, Sundolf MK, Niu Z, Peng L, Sun J, Marzinsky A, Melton GB, Tignanelli CJ*. **Development, External Validation, and Clinical Deployment of a Multimodal AI Model Toward Automated Activation of Evidence-Based Rib Fracture Care.** *Trauma Surgery & Acute Care Open*. Submitted August 2026 **[Under Review]**.

Conference Presentations

[C1] ***Thakkar V****, Silverman GM, et al.* **LLMs vs. Traditional NLP for Clinical Symptom Extraction: Comparative Analysis Across COVID-19 and Long COVID Cohorts.** *AIME 2024*, Salt Lake City, UT, May 2025. **Oral presentation.**

RESEARCH EXPERIENCE

1. **Automated Activation of Evidence-Based Rib Fracture Care Multimodal AI for Automated Activation of Evidence-Based Rib Fracture Care** (2023 – 2026)

University of Minnesota, Department of Surgery | PI: Dr. Christopher J. Tignanelli | Manuscript under review, TSACO 2026 [J3] | AHRQ-Funded

Clinical Motivation: Rib fractures occur in ~15% of trauma patients and carry 22% mortality in older adults. A proven CDS system reduces mortality by **40%** and unplanned ICU transfers by **70%**, but reached only **50% of eligible patients** because activation required manual physician initiation per encounter. The objective was to automate CDS activation by identifying rib fracture patients from routine ED data.

- Built a dual-branch multimodal model: a DenseNet-121 branch over AP chest radiographs, pretrained on natural images and fine-tuned on chest radiographs before task-specific adaptation, and a three-layer fully connected branch over 25 structured EHR variables available at ED presentation (vital sign extremes, laboratory values, age, pain score, chief complaint category), with branch outputs concatenated into a final classification layer.
- Implemented the imaging pipeline: automated quality screening, duplicate-encounter removal by database matching, CLAHE contrast normalization, and Grad-CAM activation maps for spatial attribution.
- Developed and validated across 28,126 adult patients from four U.S. academic health systems comprising 35 hospitals: internal development and validation on 21,658 patients (Sites 1–2), external validation on 6,468 patients (Sites 3–4) withheld from all model development decisions.
- Achieved AUROC 0.90 with 96% negative predictive value at Site 1 and 0.80 at Site 2 internally, and 0.70 to 0.75 on external validation, outperforming a radiograph-only model at every site; performance consistent across demographic subgroups after correction for multiple comparisons. Reported in accordance with TRIPOD+AI.
- Deployed in production at M Health Fairview in February 2026 at a prespecified Youden-optimal threshold, running six months in silent shadow mode over 16,994 ED encounters (production AUROC 0.73, 95% CI 0.67–0.79), with discrimination rising to 0.82 by two to three hours after arrival as clinical data accumulated over the encounter.

2. Longitudinal Symptom Severity Tracking in Vagus Nerve Stimulation Patients: A Two-Stage LLM-Based Pipeline with Explainable AI (2023 – 2025)

University of Minnesota, Department of Surgery | Principal Investigator (PI): Dr. Christopher J. Tignanelli | JAMIA Open 2026 [J2] | NIH U54AT01230

Clinical Motivation: VNS is used for drug-resistant epilepsy and treatment-resistant depression, but no automated system existed to track how patient symptoms changed longitudinally after device implantation. Traditional monitoring relied on patient self-report and periodic surveys, which are noisy and infrequent. The objective was to built a foundational dataset of symptom severity trajectories in VNS patients using LLMs.

- Built a two-stage LLM pipeline for longitudinal symptom severity extraction from clinical notes. Stage 1 fine-tuned LLaMA-3.2-1B, 3.2-3B and 3.3-70B to classify 13 symptoms as present, absent or negated, configured as 13 parallel 3-class classifiers with a 39-neuron output layer, training only the final classification layer with pretrained weights frozen. Stage 2 assessed severity by structured prompting with the best Stage 1 model.
- Designed the six-level ordinal symptom-burden scale with clinical collaborators (not mentioned, negated, present with severity not inferable, mild, moderate, severe) and led the annotation effort across iterative rounds with inter-rater reliability computed at each stage.
- Trained and validated on 1,427 annotated clinical notes from 95 patients across five corpora (MHFV-COVID, MHFV-PASC, N3C-COVID, NSQIP, MHFV-VNS), partitioned at patient level so no patient appeared in more than one split.
- Achieved weighted F1 0.98 for seizures and 0.99 for self-harm with LLaMA-3.3-70B and severity F1 of 0.76 to 0.96, reporting macro-averaged F1 alongside weighted F1 to avoid majority-class domination, with bootstrapped 95% confidence intervals from 1,000 resamples and equity analyses across racial and gender subgroups.
- Evaluated cross-corpus generalization, with seizure F1 of 0.98 on VNS notes versus 0.91 on NSQIP notes.
- Produced the first structured longitudinal symptom-severity dataset for VNS patients derived from routine clinical notes rather than periodic surveys, with quarterly trajectories for 23 VNS patients against a 10-patient surgical control cohort, and applied SHAP to an ordinal gradient-boosting model of inter-symptom structure.

3. LLMs versus Rule-Based Information Extraction for Patient Symptomatology (2023-2025)

University of Minnesota, Department of Surgery and NLP/Information Extraction group | PI: Dr. Christopher J. Tignanelli | PLOS One 2025 [J1]

Clinical Motivation: Patient symptoms are central to diagnosis and prognosis but are rarely captured in structured EHR data, remaining embedded in clinical notes. An existing rule-based pipeline had extracted COVID-19 and long COVID symptoms across 12 Midwest hospitals and 60 primary care clinics to monitor over 83,850 post-COVID patients, but relied on labor-intensive, expert-built lexicons. The objective was to test whether fine-tuned LLMs could match an established rule-based system for symptom extraction from clinical notes.

- Compared fine-tuned LLaMA2-13B and LLaMA3-8B against BioMedICUS, an established rule-based clinical information extraction system, across three corpora (UMN-COVID, UMN-PASC, N3C-COVID), with prevalence, keyword and fairness analyses.
- Found that BioMedICUS outperformed the fine-tuned LLMs in most cases, reaching macro-averaged F1 of 0.70 for positive mention detection on UMN-PASC.
- Documented a representation-dependence effect: using BioMedICUS header detection, symptoms under a history-of-present-illness header were identified 785 times in RTF-structured notes versus 424 times in unstructured versions of the same corpus.

4. Patient-Specific 3D Knee Reconstruction from Planar Radiographs (2020 – 2022)

TechMah Medical, Knoxville, TN | Supervisors: Dr. Emam Fatah and Dr. Mohammad Mahfouz

Clinical Motivation: Surgeons ordering implant components for total knee replacement need components sized to the patient's individual bone geometry. Standard CT scans provide 3D anatomical detail but expose patients to significant radiation and add cost and scheduling burden. The objective was to recover patient-specific 3D bone geometry from standard 2D X-ray images alone.

- Developed a modified dual-encoder-decoder U-Net for knee bone segmentation from planar radiographs, trained on 250 patients from UTK hospitals with paired CT and X-ray and ground-truth 3D surface models derived by expert segmentation.
- Designed the pose-estimation training data around a labeling bottleneck: because 3D pose ground truth for real radiographs requires expert input and does not scale, generated a synthetic training set by rendering binary projections from 316 tibia and 312 femur CAD models from the internal CT atlas, each rotated over x and z in [−15°, +15°] at 1° increments, with AP and lateral orthographic views.
- Replaced manual, operator-dependent initialization of a KPCA statistical shape model built from 300 CT-derived meshes with CNN pose estimation from binary silhouettes, reducing SSM convergence failures from approximately 15% to under 2% and initialization time from 5 to 10 minutes to under 2 seconds.
- Trained on 4× NVIDIA V100 GPUs on GCP; delivered under FDA-regulated development conditions.

5. Multimodal Retrieval and Robotic Vision (2022-2025)

Neocortex Robotics, Nashville, TN | *Supervisor: Dr. Goutham Mallapragada*

Motivation: Industrial bin-picking robots must locate and grasp individual parts from cluttered bins in which objects overlap, touch, and vary in position and orientation. Reliable picking requires separating each object at the pixel level and recognizing what it is, including parts the system was not explicitly trained on. The objective was to build vision models and a visual retrieval system to guide the robot's picks from RGB and depth images.

- Built a CLIP-based image embedding pipeline with a PostgreSQL and PGVector index and a custom LangChain retriever for multimodal retrieval over a large image corpus.
- Fine-tuned Mask R-CNN, DeepLab, PointRend and YOLO for instance segmentation, and hand-annotated 6,000 RGB and depth images for training.

ACADEMIC PROJECTS

1. **Camera Calibration and Stereo 3D Reconstruction: From Projective Geometry to Dense Point Clouds** (Aug 2018 – Dec 2019)
   Vanderbilt University, Dept. of Electrical Engineering | Advisor: Dr. Alan Peters | Master's Research Project

Implemented the complete geometric computer vision pipeline from scratch: intrinsic camera calibration, epipolar geometry estimation, stereo pose recovery, triangulation, and dense depth generation.

2. **An Empirical Investigation of CNN Design for Image Classification: Theory, Fine-Tuning, and Comparative Metrics** (Aug 2018 – Dec 2019)
   Vanderbilt University, Dept. of Electrical Engineering | Advisor: Dr. Alan Peters | Master's Research Project

Compared four canonical CNN families under identical training protocols on CIFAR-10 using macro-averaged precision, recall and F1 rather than top-1 accuracy, isolating the contributions of architectural design, loss function, optimization strategy and regularization.

RESEARCH PROPOSALS

1. **Vision-Language Models for Rib Fracture Localization, Clinical Report Generation, and Patient-Readable Summaries** (Proposed Q4 2024)
   University of Minnesota, Department of Surgery | Proposed by Vedansh Thakkar

Identified four limitations of the deployed rib fracture model: a single binary probability without indicating which rib is affected or whether it is displaced; Grad-CAM heatmaps providing rough attention rather than clinical-grade spatial precision; free-text clinical notes entirely unused despite containing mechanism of injury, pain localization and clinical suspicion; and a probability score legible to data scientists but not to patients. Proposed a **trimodal vision-language** architecture.

2. **Replacing Statistical Shape Modeling with a Generative Neural Embedding for 3D Bone Reconstruction** (Proposed May 2021)
   TechMah Medical | Proposed by Vedansh Thakkar | Independently validated: Shiode et al., Scientific Reports, July 2021

By mid-2021 the segmentation and pose-estimation CNNs had reduced SSM convergence failures from approximately 15% to under 2% and initialization time from 5 to 10 minutes to under 2 seconds, leaving the KPCA-SSM fitting stage as the dominant bottleneck: an iterative, non-convex optimization requiring 18 to 20 minutes per bone, non-differentiable and therefore preventing end-to-end training, and bounded in accuracy by 300 CT-derived training meshes. Proposed replacing it with a learned generative embedding.

PROFESSIONAL EXPERIENCE

Data Scientist *Dec 2025 – Present*
Yash Technologies Indore, MP, India
Manager: Mahendra Singh Chouhan
Building and deploying agentic AI systems at enterprise level with a team of interns, research engineers and scientists.

Artificial Intelligence Manager *Mar 2023 – Jul 2025*
Department of Surgery, University of Minnesota Remote, USA
Supervisors: Dr. Christopher J. Tignanelli (primary), Dr. Ju Sun, Dr. Nicholas Ingraham
Led two clinical AI projects involving fine-tuning LLMs and deep neural networks with deployment at M Health Fairview.

Machine Learning Engineer *Feb 2022 – Oct 2025 (Concurrent with UMN)*
Neocortex Robotics Nashville, TN, USA
Manager: Dr. Goutham Mallapragada
Built and deployed several deep learning models and a visual RAG system for guidance of industrial bin picking robots.

Machine Learning R&D Engineer *Jan 2020 – Feb 2022*
TechMah Medical Knoxville, TN, USA

Supervisors: Dr. Emam Fatah & Dr. Mohammed Mahfouz
Built and deployed deep learning models for 2D-3D bone reconstruction pipeline, including image segmentation and pose estimation.

Embedded Systems Engineer *May 2017 – July 2018*
Uni-info Telecom Services Pvt. Ltd. Indore, MP, India
Supervisors: Pranay Parwal
Built home automation systems using Arduino.

TECHNICAL SKILLS

Clinical NLP and LLMs: LLM fine-tuning (LLaMA, Mistral, Qwen, GatorTron, ClinicalBERT), prompt engineering, LoRA/QLoRA, clinical NLP (NER, symptom extraction, severity scoring), BioMedICUS, EHR/FHIR integration, Epic Systems

Medical Imaging: DICOM preprocessing, DenseNet-121, U-Net variants, BiomedCLIP, vision transformers, Detectron2, YOLO, Mask R-CNN, PointRend, CLAHE, MONAI, OpenCV, stereo 3D reconstruction, statistical shape modeling and kernel PCA, Grad-CAM

Evaluation and Statistics: macro versus weighted F1, per-class metrics and confusion matrices, bootstrapped confidence intervals, DeLong test, equalized odds and accuracy parity, inter-rater reliability (Fleiss' kappa), SHAP, TRIPOD+AI reporting

Machine Learning Frameworks: PyTorch, TensorFlow, Keras, JAX, Hugging Face Transformers, scikit-learn, XGBoost, MONAI, LangChain

Clinical Data: MIMIC-IV, MIMIC-CXR, N3C, NSQIP

Agentic AI: RAG systems, CLIP-based visual retrieval, PGVector/FAISS, agentic workflows, MCP tools, function calling

Data and Infrastructure: Databricks, PySpark, Delta Lake, Airflow, MLflow, Kubernetes/OpenShift, CI/CD, AWS, GCP, Azure, Docker, model monitoring (PSI, KL divergence)

Programming: Python, SQL, R, MATLAB, C++, Git, Linux

HONORS

- Graduated **Summa Cum Laude**, Vanderbilt University (GPA 3.93/4.00); completed M.S. five months ahead of schedule, December 2019.
- Selected for oral presentation at **AIME 2024** (Artificial Intelligence in Medicine), Salt Lake City, UT, May 2025.

LEADERSHIP, SERVICE, AND MENTORING

- **Peer Reviewer** | JAMIA Open 2025
- **Annotation Team Lead** | University of Minnesota NIH funded Project (2023 – 2025).
- Recruited, trained, and coordinated 6 clinical SMEs (2 attending physicians, 2 residents, 2 medical students); managed 3 iterative annotation rounds, IRR calibration, and guideline refinement over 8 to 10 months.
- **AI Infrastructure Lead** | University of Minnesota Department of Surgery (2023 – 2025).
- Built AI development infrastructure from scratch in a department with no prior data science presence: established project timelines, coordinating clinical collaborators, managing multi-institutional data agreements across 5 sites.
- Mentored 2 undergrad (BS and BA) and 1 graduate student (MD) at the University of Minnesota.